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TREND-BASED MOCK PAPER

GATE Geology and Geophysics

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Geology and Geophysics
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Choose similar to 'volcanic':

- (A) Gentle
- (B) Explosive
- (C) Calm
- (D) Peaceful

Section B: Geology and Geophysics

Q2. [1-NAT] [1 mark]

Mean of 1,2,3,4,5:

(Numerical Answer Type — Enter the numerical value)

Q3. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q4. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q5. [1-MCQ] [1 mark]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q6. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q7. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q8. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q9. [1-MCQ] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)}(1+z)$

Q10. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q11. [1-NAT] [1 mark]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q12. [1-MCQ] [1 mark]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4

(D) 1

Q13. [1-MCQ] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q14. [1-MCQ] [1 mark]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q15. [1-MCQ] [1 mark]

The Laplace transform of $t e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q16. [1-NAT] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q17. [1-MCQ] [1 mark]

$P(|Z| < 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q18. [1-NAT] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q19. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q20. [1-NAT] [1 mark]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q21. [2-NAT] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q22. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q23. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
 - (B) Positive definite
 - (C) Diagonally dominant
 - (D) Non-singular
-

Q24. [2-MSQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
 - (B) $\|u\| + \|v\|$
 - (C) $\|u\| / \|v\|$
 - (D) $\sqrt{\|u\| * \|v\|}$
-

Q25. [2-NAT] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\operatorname{trace}(A^2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q26. [2-MSQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
 - (B) $y = x \ln(x)$
 - (C) $y = x \sin(\ln x)$
 - (D) $y = x^2$
-

Q27. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ equals:

- (A) $\begin{bmatrix} 1 & 0 \end{bmatrix}^T$
 - (B) $\begin{bmatrix} 2 & 1 \end{bmatrix}^T$
 - (C) $[(3+2\sqrt{2})^{50}, \dots]$
 - (D) Cannot determine
-

Q28. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
 - (B) 0.5
 - (C) 0.6
 - (D) 0.4
-

Q29. [2-NAT] [2 marks]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q30. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q31. [2-MCQ] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{(\lambda(z-1))}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)}(1+z)$

Q32. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q33. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q34. [2-MSQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q35. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q36. [2-NAT] [2 marks]

The differential equation $y dx - x dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q37. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q38. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q39. [2-MCQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q40. [2-NAT] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q41. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q42. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q43. [2-MCQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q44. [1-MCQ] [1 mark]

Hardest mineral:

- (A) Quartz
- (B) Diamond
- (C) Topaz
- (D) Corundum

Q45. [1-MCQ] [1 mark]

Study of fossils:

- (A) Paleontology
- (B) Petrology
- (C) Mineralogy

(D) None

Q46. [1-MCQ] [1 mark]

Law of superposition by:

- (A) Steno
- (B) Lyell
- (C) Darwin
- (D) Hutton

Q47. [1-MCQ] [1 mark]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q48. [1-NAT] [1 mark]

Hardest mineral on Mohs scale:

(Numerical Answer Type — Enter the numerical value)

Q49. [2-MCQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q50. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q51. [2-MCQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q52. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q53. [2-MCQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q54. [2-NAT] [2 marks]

Hardest mineral on Mohs scale:

(Numerical Answer Type — Enter the numerical value)

Q55. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q56. [2-NAT] [2 marks]

Igneous rock from lava:

(Numerical Answer Type — Enter the numerical value)

Q57. [2-MSQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q58. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q59. [2-NAT] [2 marks]

Hardest mineral on Mohs scale:

(Numerical Answer Type — Enter the numerical value)

Q60. [2-NAT] [2 marks]

Igneous rock from lava:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

Gravity anomaly due to:

- (A) Density variation
- (B) Magnetism
- (C) Electricity
- (D) None

Q62. [1-MCQ] [1 mark]

P-waves are:

- (A) Primary
- (B) Secondary
- (C) Surface
- (D) Love

Q63. [1-MCQ] [1 mark]

Earth's magnetic field generated by:

- (A) Core
- (B) Crust
- (C) Mantle
- (D) Atmosphere

Q64. [1-NAT] [1 mark]

Landsat uses:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Igneous rock from:

- (A) Sediment
 - (B) Cooling magma
 - (C) Metamorphism
 - (D) Organic
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1NAT	B	1
3	1MCQ	B	1
4	1MCQ	A	1
5	1MCQ	A	1
6	1MCQ	A	1
7	1MCQ	A	1
8	1MCQ	A	1
9	1NAT	A	1
10	1MCQ	B	1
11	1MCQ	B	1
12	1MCQ	C	1
13	1NAT	B	1

14	1MCQ	C	1
15	1MCQ	A	1
16	1MCQ	B	1
17	1MCQ	C	1
18	1MCQ	B	1
19	1MCQ	A	1
20	1MCQ	A	1
21	1NAT	A	1
22	1MCQ	C	1
23	1MCQ	B	1
24	1MCQ	A	1
25	1MCQ	A	1
26	1NAT	B	1
27	1MCQ	A	1
28	1NAT	A	1

29	1MCQ	B	1
30	1NAT	B	1
31	2NAT	A	2
32	2NAT	A	2
33	2MCQ	C	2
34	2MSQ	A	2
35	2NAT	A	2
36	2MSQ	A	2
37	2MCQ	A	2
38	2MCQ	B	2
39	2MCQ	B	2
40	2MCQ	B	2
41	2NAT	B	2
42	2MCQ	A	2
43	2MCQ	B	2

44	2MCQ	A	2
45	2MCQ	B	2
46	2MCQ	A	2
47	2NAT	A	2
48	2MSQ	A	2
49	2MCQ	B	2
50	2MCQ	B	2
51	2NAT	B	2
52	2NAT	B	2
53	2MCQ	B	2
54	2MCQ	A	2
55	2MCQ	B	2
56	2MCQ	B	2
57	2NAT	B	2
58	2MSQ	B	2

59	2NAT	A	2
60	2MCQ	B	2
61	2MCQ	C	2
62	2MCQ	B	2
63	2NAT	B	2
64	2NAT	B	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: difficult

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Statistics | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Geology | Topic: Minerals | Difficulty: easy

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Geology | Topic: Fossils | Difficulty: difficult

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Geology | Topic: Stratigraphy | Difficulty: moderate

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Geophysics | Topic: Gravity | Difficulty: moderate

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Geophysics | Topic: Seismic | Difficulty: moderate

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Geophysics | Topic: Magnetism | Difficulty: moderate

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Remote Sensing & GIS | Topic: Satellite | Difficulty: moderate

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Petrology | Topic: Rocks | Difficulty: easy

Q11.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q12.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q13.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q14.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q15.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q16.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q17.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q18.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q20.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q22.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q24.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q25.

Answer: A

Solution:

$L\{f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q27.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q28.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q29.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4}$ approx 0.908.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}*[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: easy

Q39.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q40.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: easy

Q41.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q42.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: easy

Q44.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q45.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q46.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q47.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q50.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q51.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: easy

Q52.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q53.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q54.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q55.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q56.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q58.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q59.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q60.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q61.

Answer: C

Solution:

Rolle's: f continuous on $[a, b]$, differentiable on (a, b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a, b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q62.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C * x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C * x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C * x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q63.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q64.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q65.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 42 questions, 65 marks
Geology: 17 questions, 29 marks
Geophysics: 3 questions, 3 marks
General Aptitude: 1 questions, 1 marks
Remote Sensing & GIS: 1 questions, 1 marks
Petrology: 1 questions, 1 marks

Difficulty Distribution:

Easy: 22 | Moderate: 36 | Difficult: 7

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.